

Sri Lanka Institute of Information Technology



Kandy Uni

IT2011 - Artificial Intelligence and Machine Learning

Lab Submission

Worksheet No: 02

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<Batch 2>

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Exercise 01- Part1

PartyRock

GAIF Activities - Structuring an Effective Prompt

Introduction

- This activity has two parts. Scroll through the instructions to find and follow each step for parts I and II.
- In part I of the activity you refine a prompt by continuing to add elements to the initial prompt.
- In part II you use a restructured version of the prompt to help the model better understand the intent. Complete this step after reviewing the related topics in your student guide.

Instructions

- For each numbered prompt element, copy and paste the example into the Activity 1 Prompt Box and run (play) the app. Review the results in the Response 1 widgets.
- Continue to add each element and run the app after each addition.
- Share your observations
 - How do the model responses differ?
 - How are they the same?
 - How does each additional prompt element change the responses?

Describing the tasks the model should perform

1. Task description:

Write a blog post about the potential health impacts of plant-based diets

2. Step-by-step instructions:

Activity 1 Prompt Box

Write a blog post about the potential health impacts of plant-based diets.

Response 1 using Amazon Nova Pro

Response 1 using Claude Haiku

Response 1 using Claude Sonnet 3.5

Figure 1:Part1ex1- prompt

PartyRock

GAIF Activities - Structuring an Effective Prompt

Response 1 using Amazon Nova Pro

The Potential Health Impacts of Plant-Based Diets

In recent years, plant-based diets have surged in popularity, driven by growing awareness of their potential health benefits, environmental sustainability, and ethical considerations. But what does the science say about the health impacts of adopting a plant-based diet? Let's dive into the research and explore the potential advantages-and a few considerations-of making plants the centerpiece of your plate.

What is a Plant-Based Diet?

A plant-based diet emphasizes foods derived

Response 1 using Claude Haiku

Here's a draft blog post on the health impacts of plant-based diets:

The Health Benefits of Plant-Based Eating: What Science Says

Plant-based diets have gained significant popularity in recent years, and for good reason. More than just a dietary trend, mounting scientific evidence suggests that plant-forward eating patterns can offer substantial health benefits. Let's explore the potential positive impacts of embracing more plant-based foods.

Cardiovascular Health One of the most well-documented benefits of plant-based diets is

Response 1 using Claude Sonnet 3.5

Here's a blog post about the health impacts of plant-based diets:

The Health Benefits and Considerations of Plant-Based Diets

In recent years, plant-based diets have gained significant popularity, with more people choosing to reduce or eliminate animal products from their meals. But what does the science say about the health impacts of following a plant-based diet? Let's explore the potential benefits and considerations.

Key Health Benefits

• **Heart Health** Plant-based diets are associated with a lower risk of heart disease, likely due to their high fiber and low saturated fat content.

Return to main activities app

Figure 2:Part1ex1-output

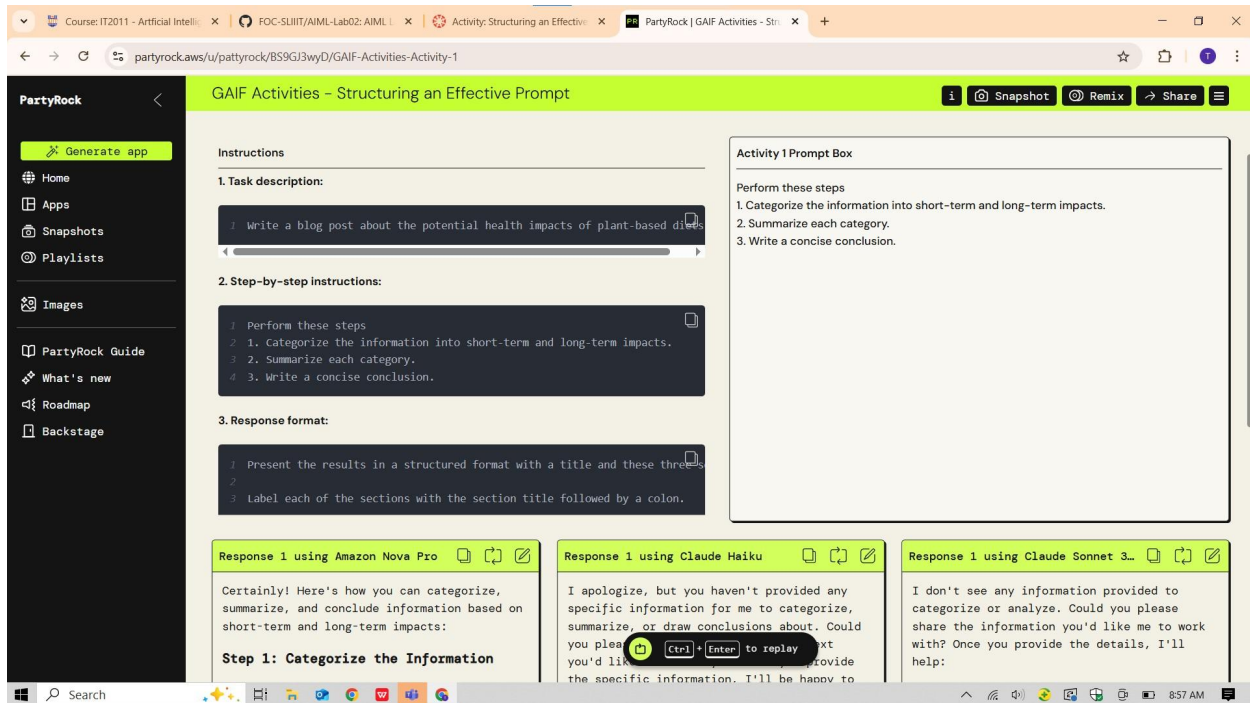


Figure 3:Part1ex2- prompt

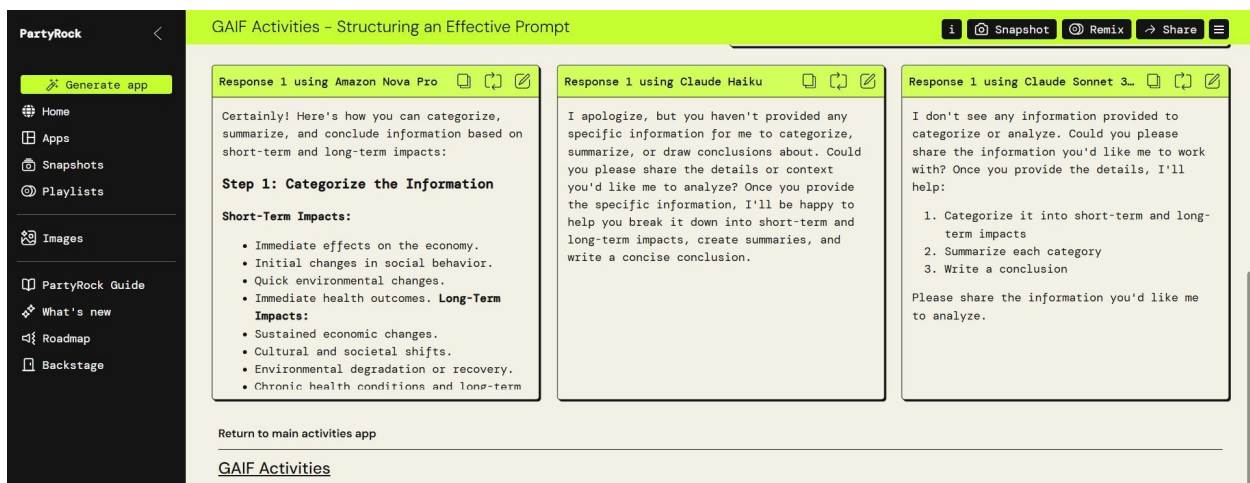


Figure 4:Part1ex2- output

Exercise1- Part 2

Instructions

Part II: Using a tagged structure.

Complete part II after reviewing the course materials related to how to organize and tag prompt elements.

- Clear the Activity 1 Prompt Box.
- Copy and paste the restructured prompt into the prompt box and run (play) the app.
- Share your observations: Do you notice any difference in how the model's provided the answer as compared to the results for part I?

Restructured prompt

```
1 <tone>
2 - role: High-energy TV chef who is passionate about food education.
3 - audience: teenagers.
4 </tone>
5 <context> The goal of the blog post is to promote a balanced view of plant-based diets
6 </context>
7 <instructions>
8 - Categorize the information into short-term and long-term impacts.
9 - Summarize each category.
10 - Write a concise conclusion.
11 </instructions>
12 <constraints>
13 - Do not discuss weight.
14 - Do not reference in the output that the audience is teenagers.
15 </constraints>
16 <response format> Include a title. Include these three sections: Introduction, Body, Conclusion. Label each of the sections with the section title followed by a colon.
17 </response format>
18 <task> Using the role and audience described in <tone> and the context in <context>,
19 write a blog post about the health impacts of plant-based eating.
```

Activity 1 Prompt Box

```
<tone>
- role: High-energy TV chef who is passionate about food education.
- audience: teenagers.
</tone>
<context> The goal of the blog post is to promote a balanced view of plant-based diets
</context>
<instructions>
- Categorize the information into short-term and long-term impacts.
- Summarize each category.
- Write a concise conclusion.
</instructions>
<constraints>
- Do not discuss weight.
- Do not reference in the output that the audience is teenagers.
</constraints>
<response format> Include a title. Include these three sections: Introduction, Body, Conclusion. Label each of the sections with the section title followed by a colon.
</response format>
<task> Using the role and audience described in <tone> and the context in <context>,
write a blog post about the health impacts of plant-based eating.
```

Figure 5: part2 prompt

Response 1 using Amazon Nova Pro

Title: The Vibrant World of Plant-Based Diets: Uncovering Health Impacts

Introduction: Hey there, food enthusiasts! Today, we're diving into the exciting world of plant-based diets and exploring their incredible health impacts. Whether you're a veggie lover or just curious about adding more greens to your plate, this post is for you. Let's uncover the short-term and long-term benefits of embracing a plant-based lifestyle!

Body:

Short-Term Impacts: When you start incorporating more plant-based foods into

Response 1 using Claude Haiku

Plant Power: Exploring the Health Impacts of Plant-Based Eating

Introduction: Hey food adventurers! Let's dive into the world of plant-based diets and unpack what's REALLY going on with this epic eating style. Buckle up for a flavor-packed journey through nutrition science!

Body: Short-Term Health Impacts: • Increased energy levels • Improved digestive function • Enhanced nutrient absorption • Potential reduction in inflammation Summary: Switching to plant-based eating can give your body an immediate nutrition boost, helping you feel more energized and balanced.

Response 1 using Claude Sonnet 3.5

The Real Deal About Plant-Based Eating: What Happens When You Go Green!

Introduction: BAM! Let's kick things into high gear and talk about something super exciting - plant-based eating! *waves spatula enthusiastically* You know what's amazing? When you fuel your body with plants, it's like giving your engine a complete upgrade! Let me break down what happens when you make plants the star of your plate!

Body: SHORT-TERM IMPACTS - INSTANT VICTORIES! 🌟

- KAPOW! More energy throughout the day
- BOOM! Better digestion (hello, fiber!)

Figure 6: part2 response

This exercise is related to the way how each model differs for the same prompt but with different models. When comparing the responses that I got for these, I can say the 3 models are different to one another. I got responses from,

Response 1: Using Amazon Nova Pro

Response 2: Using Claude Haiku

Response 3: Using Claude Sonnet

I got the lengthiest response and the bold texts from Amazon Nova Pro compared to the other 2 models.

Exercise 02

The screenshot shows the LLM Board interface with various filters and a table of models. The filters include Frontiers Models, Open Weights, Size Class, Reasoning, and Model Status. The table lists models like Grok 4, o3-pro, o3, o4-mini (high), Gemini 2.5 Pro, Qwen3 235B 2507 (Reasoning), DeepSeek R1 0528, Grok 3 mini Reasoning (high), Claude 4, and Opus. The table columns include Model, Creator, Context Window, Artificial Analysis Intelligence Index, IFBench (Instruction Following), AIME 2025 (Competition Math), Blended Price (USD/1M Tokens), Median Output Tokens/s, and Latency.

FEATURES		INTELLIGENCE				PRICE	OUTPUT	LATENCY
MODEL	CREATOR	CONTEXT WINDOW	ARTIFICIAL ANALYSIS INTELLIGENCE INDEX	IFBENCH (INSTRUCTION FOLLOWING)	AIME 2025 (COMPETITION MATH)	BLENDED USD/1M Tokens	MEDIAN Tokens/s	MED First Ch
Grok 4	X	256k	67	54%	93%	\$6.00	62.3	10
o3-pro	OpenAI	200k	67			\$35.00	32.7	8
o3	OpenAI	200k	67	71%	88%	\$3.50	190.0	1
o4-mini (high)	OpenAI	200k	66	69%	91%	\$1.93	111.7	4
Gemini 2.5 Pro	Google	1m	64	49%	88%	\$3.44	147.3	3
Qwen3 235B 2507 (Reasoning)	Alibaba	256k	63	51%	91%	\$2.63	63.5	1
DeepSeek R1 0528	deepseek	128k	59	40%	76%	\$0.96	23.5	3
Grok 3 mini Reasoning (high)	X	1m	59	46%	85%	\$0.35	208.6	0
Claude 4	Anthropic	200k	58	54%	73%	\$30.00	49.4	1
Opus	OpenAI	200k	58	54%	73%	\$30.00	49.4	1

Figure 7: LLM Board

Large Language Models (LLMs) are advanced artificial intelligence systems designed to understand and generate human-like text. In this image popular LLMs can be seen.

Favourite LLM rank: DeepSeek - ranked 7

Q; Try out at least two models that you haven't tried before (you will have to do a separate google search to find these products)

Models that I chose- Claude and Gemini

Explain Photosynthesis process

Photosynthesis is the process by which plants, algae, and some bacteria convert light energy (usually from sunlight) into chemical energy in the form of glucose, while simultaneously producing oxygen as a byproduct. This process is fundamental to life on Earth, as it provides the energy source for most living organisms and produces the oxygen we breathe.

The Overall Equation

The simplified chemical equation for photosynthesis is:



This means six molecules of carbon dioxide plus six molecules of water, with the help of light energy, produce one molecule of glucose and six molecules of oxygen.

The Two Main Stages

Light-Dependent Reactions (Photo Reactions) These occur in the thylakoid membranes of chloroplasts. Chlorophyll and other pigments absorb light energy, which is used to split water molecules, releasing oxygen as a waste product. This process also generates ATP (adenosine triphosphate) and NADPH, which are energy-carrying molecules that will be used in the next stage.

Light-Independent Reactions (Calvin Cycle) These take place in the stroma of chloroplasts and don't directly require light. Using the ATP and NADPH produced in the first stage, along

Reply to Claude...



Claude Sonnet 4 ▾



Figure 8: Claude ai

explain the growth of a butterfly

- ◆ The growth of a butterfly is a fascinating process called complete metamorphosis, which involves four distinct stages: egg, larva (caterpillar), pupa (chrysalis), and adult. 

Stage 1: Egg

The life cycle begins when an adult female butterfly lays tiny eggs on a plant, typically one that the future caterpillar will eat. The eggs can be round, oval, or cylindrical, and their color and texture depend on the species. They are usually attached to the plant with a glue-like substance. This stage can last for a few days to a few weeks, depending on the species and temperature.



Stage 2: Larva (Caterpillar)

When the egg hatches, a larva, or caterpillar, emerges. The caterpillar's primary job is to eat and grow. It is a voracious eater, and its first meal is often its own eggshell. It then begins to consume the plant on which it was born. As the caterpillar grows, its skin (exoskeleton) doesn't stretch, so

Ask Gemini



Deep Research



Canvas



Image



Gemini can make mistakes, so double-check it

Figure 9: Gemini ai

Exercise 03

How you can use generative AI as a university student to learn subjects like probability and statistics, data base management systems and software engineering



As a university student, you can leverage **generative AI** to enhance your learning in subjects like **Probability & Statistics, Database Management Systems (DBMS), and Software Engineering**. Here's how:

1. Probability & Statistics

Generative AI can help you:

- **Explain Concepts**

- Ask AI to simplify complex topics (e.g., Bayes' Theorem, Normal Distribution) with examples.
- Example prompt:

| "Explain conditional probability with a real-world example."

- **Solve Problems Step-by-Step**

- Use AI to break down problems (e.g., hypothesis testing, regression analysis).
- Example prompt:

| "**Solve this probability problem step-by-step: If $P(A)=0.4$, $P(B)=0.3$, and $P(A \cap B)=0.1$, find $P(A|B)$."*

- **Generate Practice Questions**

- Ask AI to create quizzes or exam-style questions.



Message DeepSeek

DeepThink (R1)

Search



AI-generated, for reference only

Figure 10: Using DeepThing R1

How you can use generative AI as a university student to learn subjects like probability and statistics, data base management systems and software engineering

Qwen3-235B-A22B-2507

Generative AI can be a powerful learning companion for university students studying complex subjects like **Probability and Statistics**, **Database Management Systems (DBMS)**, and **Software Engineering**. Here's how you can use it effectively and responsibly across these subjects:

1. Probability and Statistics

How to Use Generative AI:

- **Explain Concepts in Simple Terms**

Ask AI to break down complex ideas (e.g., Bayes' Theorem, Central Limit Theorem, hypothesis testing) into intuitive explanations with real-life examples.

Example prompt:

↓
"Explain the Central Limit Theorem like I'm a beginner, using an example from everyday life."

 Web Dev  Image Generation  Video Generation  Artifacts

How can I help you today?

  Thinking |   Search

By messaging Qwen, you agree to our [Terms of Service](#) and have read our [Privacy Policy](#).
AI-generated content may not be accurate.

Figure 11:Using Thinking

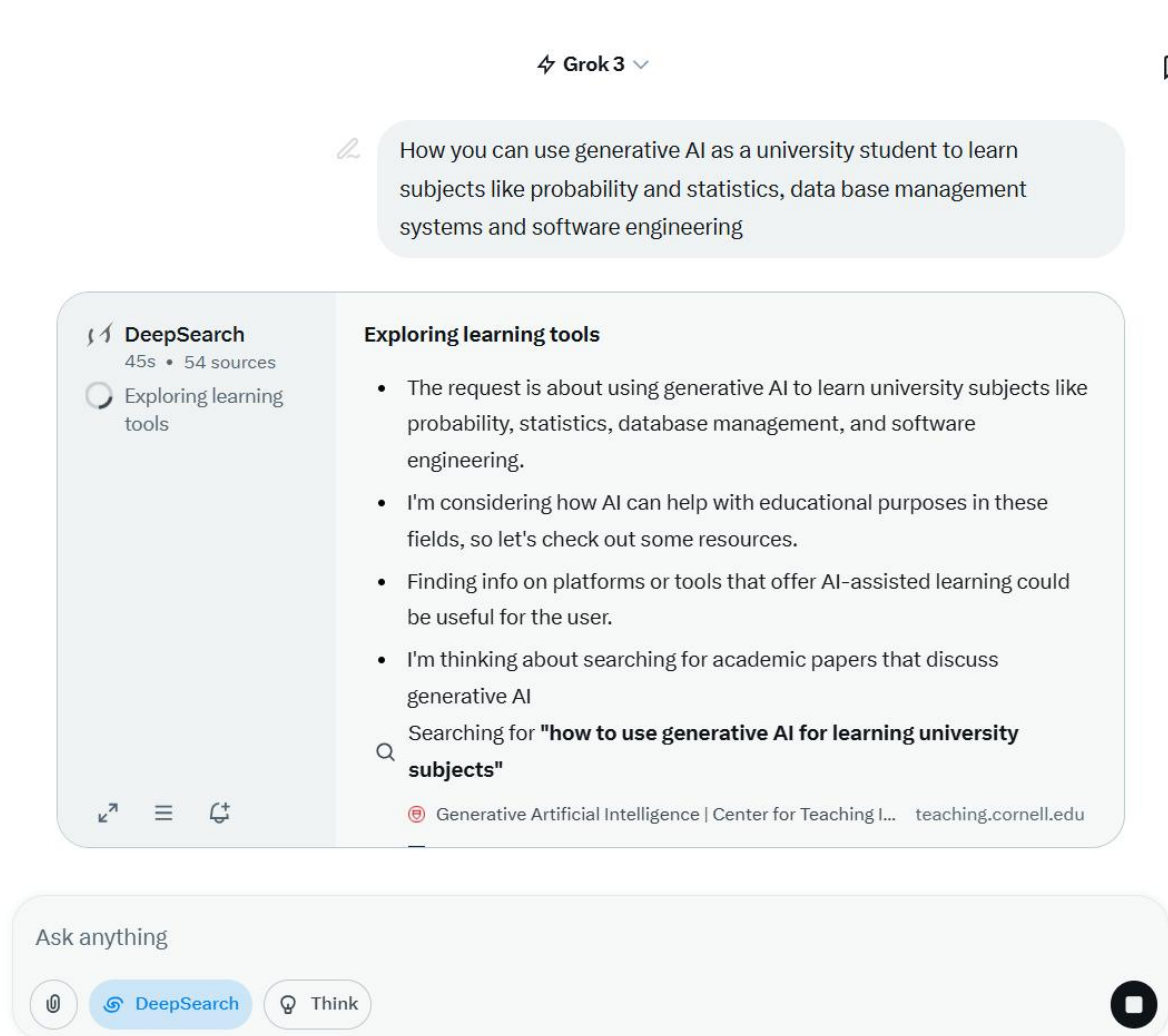


Figure 12: Using DeepSearch

Exercise 04

DeepLearning.AI

jupyter L1_student Python 3 (ipykernel) Logout

Not Trusted

File Edit View Insert Cell Kernel Help

Run

```
In [2]: from prompt_toolkit import PromptSession
from prompt_toolkit.history import InMemoryHistory
from openai import OpenAI

system_prompt = {"role": "system", "content": "You are a helpful assistant."}
user_prompt = {"role": "user", "content": "Explain how black holes are formed."}

client = OpenAI(api_key="sk-3.5-turbo")
response = client.chat.completions.create(model="gpt-3.5-turbo", messages=[system_prompt, user_prompt])
print(response.choices[0].message.content)
```

Black holes are formed when a massive star reaches the end of its life cycle and undergoes a supernova explosion. During the supernova, the outer layers of the star are blasted away into space, leaving behind the core of the star, which collapses under its own gravity.

Figure 13: Pasting the codeblock

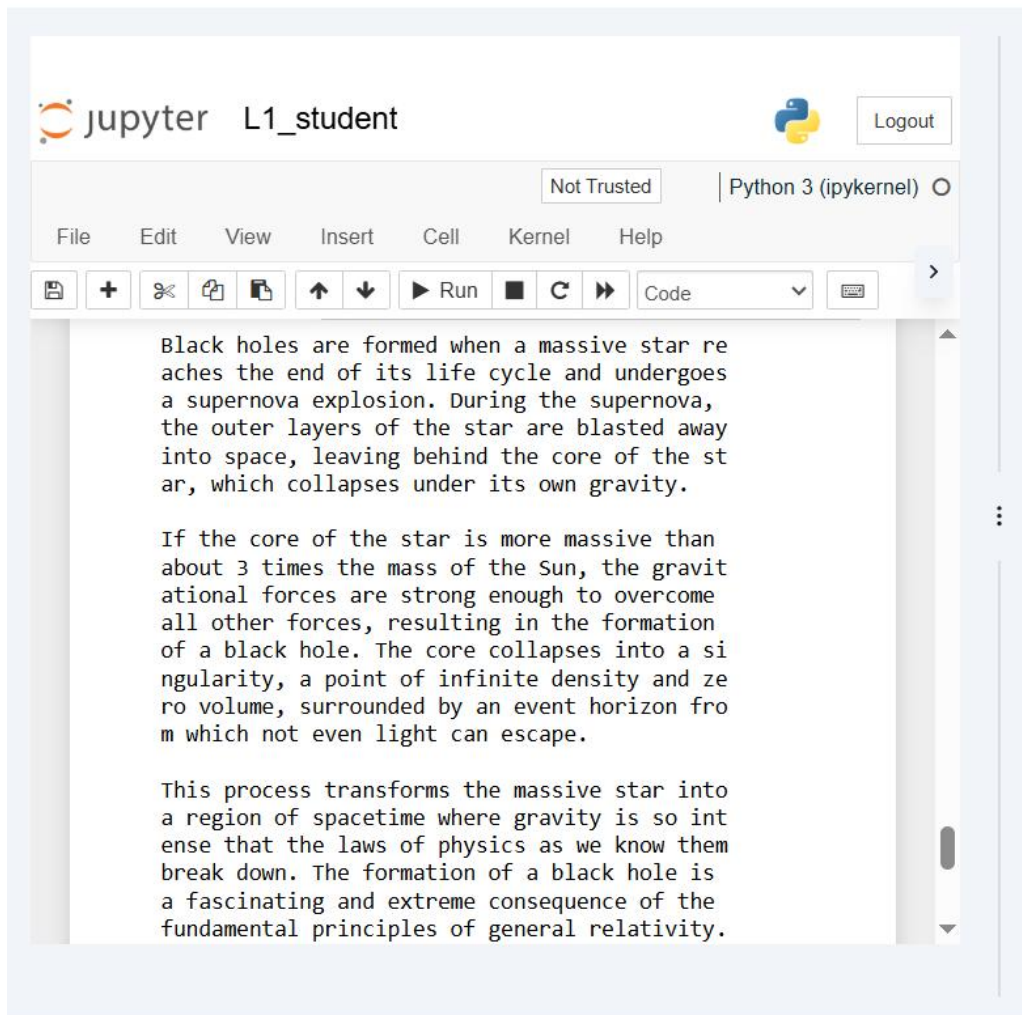


Figure 14: Output from the codeblock

Planning my Friends Birthday Party.

```
In [4]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are an expert event pla

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "I need to plan a beach-themed 2

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

Planning a beach-themed birthday party on a \$1500 budget for 30 guests can be a fun and memorable experience. Let's break down the key elements of the event:

****Venue Options:****

1. ****Public Beach:**** Consider hosting the party at a local public beach if permit requirements are met. This option could save on venue costs but may come with limitations on setup and privacy.
2. ****Backyard Beach Setup:**** Transforming a backyard into a beach paradise can be cost-effective. Utilize sand, beach umbrellas, and inflatable palm trees to create the ambiance.
3. ****Community Center with Pool:**** Renting a community center with a pool can provide a beach-like setting without the actual beach. Check for availability and any additional fees.

Figure 15: Give a User Prompt asking for advice how to plan for an event you have to do, maybe your friends Birthday Party, or a Batch Outing

2. ****Tropical Fruit Platter:**** Serve fresh fruit platters with pineapple, watermelon, mango, and berries for a refreshing touch.
3. ****Beach Snacks:**** Offer beach-themed snacks like popcorn, pretzels, and seafood appetizers. You can also include a make-your-own ice cream sundae station for dessert.

****Entertainment Activities:****

1. ****Beach Volleyball:**** Set up a beach volleyball net for guests to enjoy friendly matches.
2. ****DIY Photo Booth:**** Create a DIY photo booth with beach props for guests to capture memories.
3. ****Lawn Games:**** Include activities like cornhole, ladder toss, or beach-themed scavenger hunts for added fun.
4. ****DIY Tiki Bar:**** Create a DIY tiki bar serving tropical mocktails, such as piña coladas or daiquiris.

****Budget Tips:****

1. ****DIY Decor:**** Utilize items like beach balls, seashells, and string lights for decor. Dollar stores and thrift shops can be great sources for affordable decorations.
2. ****BYOB Policy:**** Implement a "Bring Your Own Booze" policy to cut down on beverage costs.
3. ****Digital Invitations:**** Use online platforms to send digital invitations, saving on printing and postage fees.
4. ****Go Potluck:**** Encourage guests to bring a dish to share, reducing the catering costs.

By combining these venue options, food ideas, entertainment activities, and budget-saving tips, you can create a beach-themed birthday party that will be a hit with your guests while staying within your \$1500 budget.

Figure 16: Output taken

Exercise 05

Wan 2.2 TI2V 5B

generate high quality videos using **Wan 2.2 5B Text-Image-to-Video model**,[\[model\]](#),[\[paper\]](#)

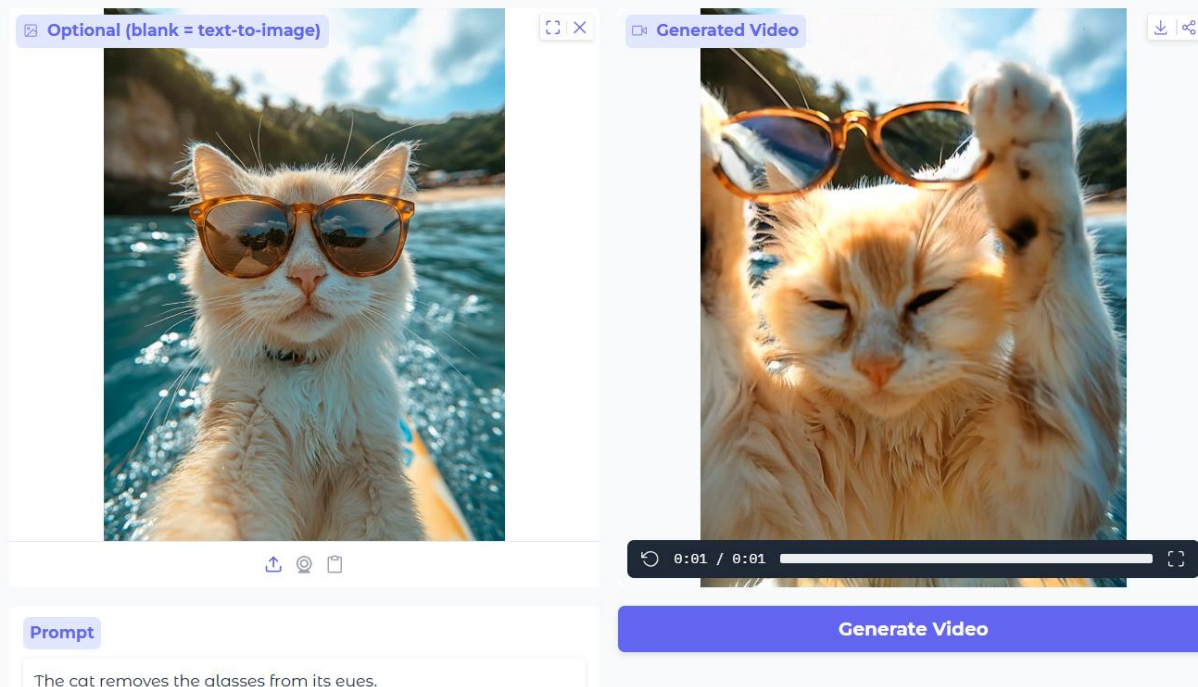


Figure 17: Generating videos from hugging face

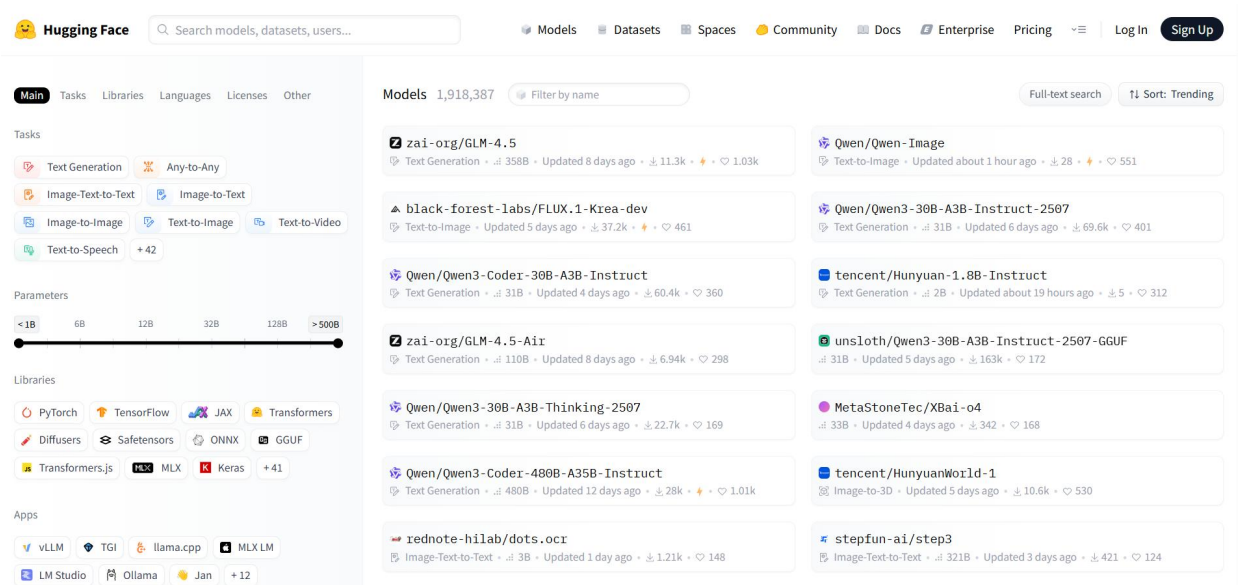


Figure 18: Models in hugging face

GLM-4.5



👉 Join our [Discord](#) community.

📖 Check out the GLM-4.5 [technical blog](#).

📌 Use GLM-4.5 API services on [Z.ai API Platform \(Global\)](#) or [Zhipu AI Open Platform \(Mainland China\)](#).

👉 One click to [GLM-4.5](#).

Model Introduction

The **GLM-4.5** series models are foundation models designed for intelligent agents. GLM-4.5 has **355** billion total parameters with **32** billion active parameters, while GLM-4.5-Air adopts a more compact design with **106** billion total parameters and **12** billion active parameters. GLM-4.5 models unify reasoning, coding, and intelligent agent capabilities to meet the complex demands of intelligent agent applications.

Both GLM-4.5 and GLM-4.5-Air are hybrid reasoning models that provide two modes: thinking mode for complex reasoning and tool usage, and non-thinking mode for immediate responses.

Figure 19:Text Generation ai model GLM

Q: Run the summarization example in a codeblock, here the Hugging Face open model will be downloaded to Google Colab and then it will be executed. Because of this it will take some time to run the model for the first time.

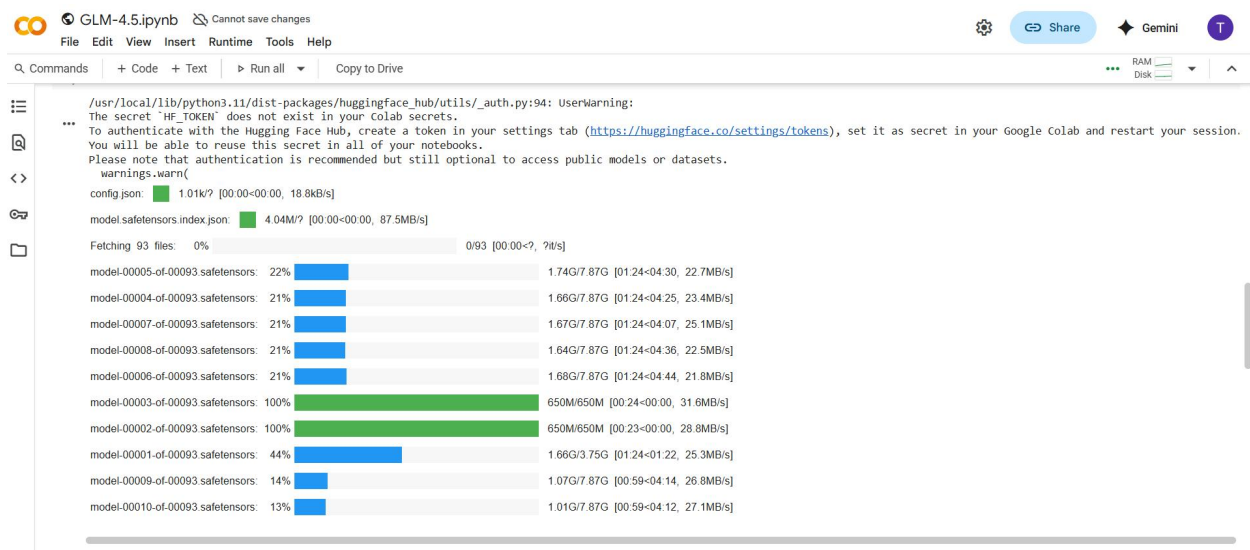


Figure 20: Running textual at google colab

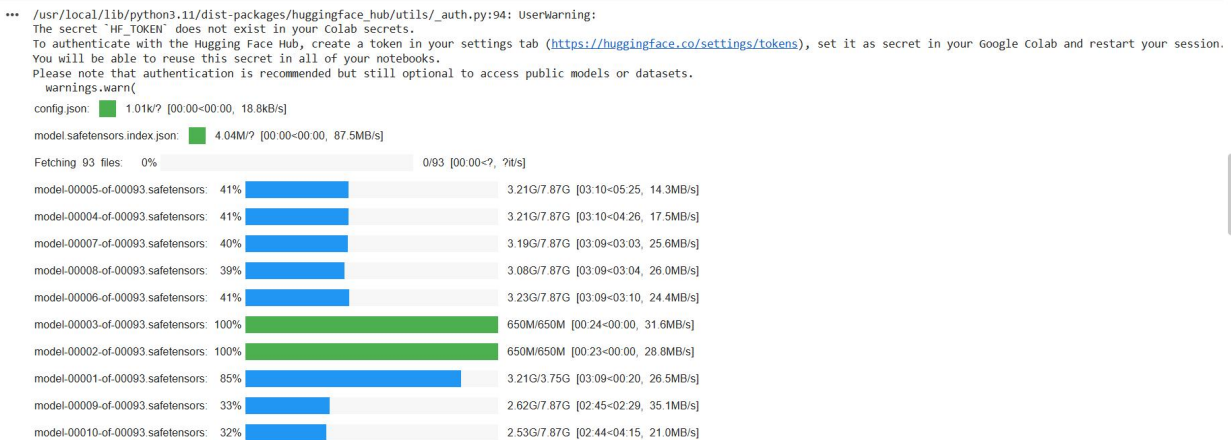


Figure 21: Running status of summarization text